Exact lens-space spectral-flow certificates

Luqman Mahmood*

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1 Exact lens-space spectral-flow certificates

This supplement provides exact finite reference data for the rank-one spectral-flow control used in the main text and tabulates the integer-spin eta formula through rank eight. Its purpose is reproducibility: it supplies no additional physical interpretation of the higher-rank invariants.

1.1 Rank-one certificate

Let p be odd, let $\gcd(p,q)=1$, and give

$$L(p,q)=S^3/((z_1,z_2)\sim(e^{2\pi i/p}z_1,e^{2\pi iq/p}z_2))$$

the complex quotient orientation. Write $q^*\in\{1,\dots,p-1\}$ for the least positive inverse of q modulo p. For $1\le n\le(p-1)/2$, define

$$R_m(p,q)=\frac{1}{p}\sum_{k=1}^{p-1}\cos\!\left(\frac{2\pi mk}{p}\right)\cot\!\left(\frac{\pi k}{p}\right)\cot\!\left(\frac{\pi qk}{p}\right). \tag{1}$$

Jeffrey's Proposition 5.2 [Jef92], in these conventions, gives the integer spectral flow

$$I_n=\frac{8q^*n^2}{p}-2+R_0(p,q)-R_{2n}(p,q). \tag{2}$$

The same proposition supplies the independent counting congruence

$$I_n\equiv 1+2\#\{\,0<j<nq^*:jq\equiv\pm n\pmod p\,\}\pmod 4. \tag{3}$$

Equivalently, the residue is 1 when $\langle q^*n\rangle_p<p/2$ and 3 when $\langle q^*n\rangle_p>p/2$.

The rank-one eta identity is

$$R_0(p,q)=4s(q,p)=-\eta_1(L(p,q)). \tag{4}$$

Thus (2) is an established rank-one control on the orientation and normalization used in the main theorem. It is not a premise of that theorem.

For exact generation, the cotangent expression is evaluated through the finite Fourier identity

$$R_m(p,q)=4\sum_{r=0}^{p-1}\left(\!\!\left(\frac{r}{p}\right)\!\!\right)\left(\!\!\left(\frac{qr+m}{p}\right)\!\!\right), \tag{5}$$

where the sawtooth is zero at integers and equals $x-\lfloor x\rfloor-\frac{1}{2}$ otherwise. Equation (5) uses rational arithmetic throughout and avoids numerical reconstruction of (1).

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1.2 Dataset and verification hierarchy

The accompanying files contain every odd $3 \le p \le 61$, every unit $q \bmod p$, and every $1 \le n \le (p-1)/2$, together with the $q = 1$ blocks at $p = 67, 97, 137$. The resulting 16,177 rows record the exact quantities

$$I_n, \quad I_n \bmod 4, \quad q^* n^2/p \bmod 1, \quad R_0, \quad R_{2n}, \quad s(q,p),$$

while each (p,q) block records $\eta_J(L(p,q))$ for $1 \le J \le 8$.

The checks have three distinct strengths.

1. **Primary acceptance test: exact row equality.** A computed crossing count passes only if it equals the certified integer I_n. A floating implementation instead compares directly with that integer using a declared tolerance.

2. **Analytic-assembly diagnostic: integrality.** The right side of (2) must be integral. This tests cancellation among independently evaluated rational terms, but it cannot detect an erroneous result that is already an integer. In particular, integrality alone does not certify a crossing count against missed or double-counted crossings.

3. **Convention diagnostic: parity.** Equation (3) is recomputed by direct enumeration, independently of the R_m assembly. A disagreement diagnoses a modulo-four sign or convention error. Agreement is weaker than exact equality, since parity cannot see errors in $4\mathbb{Z}$.

The release verifier checks the complete JSON schema and coverage, modular inverses, Dedekind and R_m values, Chern–Simons residues, spectral flows, parities, all $J = 1, \dots, 8$ theorem evaluations, duplicate freedom, CSV/JSON agreement, and every hash recorded in the manifest.

1.3 Scope boundary

Only the $J = 1$ spectral-flow control in (2) is attributed to Jeffrey. The $J \ge 2$ entries are exact evaluations of the theorem proved in the main text; their inclusion makes that theorem directly reproducible but does not independently prove it. They are compact-manifold spectral invariants, not established higher-spin Chern–Simons, one-loop, or partition-function phases. No ghost, determinant, framing, counterterm, coupling, or physical-background analysis is supplied here. Orientation must be fixed before any signed comparison.

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References

[Jef92] Lisa C. Jeffrey. Chern–Simons–Witten invariants of lens spaces and torus bundles, and the semiclassical approximation. *Communications in Mathematical Physics*, 147(3):563–604, 1992. doi:10.1007/BF02097243.